



Course Information

Course Code: CS-11303

Credit Hours: 4 (3, 1)

Lab Hours/Week: 3

Course Title: **Operating System**

Lecture Hours/Week: 3

Pre-Requisites: **None**

Design skills /techniques practiced_ Operating System _Lab

Students practiced:

1. Process Control

- Creating child processes using fork()
- Process synchronization using wait()
- Background & foreground process control

2. Shell & Command Automation

- Writing Bash scripts
- Automating repetitive tasks
- Cron job scheduling

3. File Permission & Security Implementation

- Implementing user-based access control
- Designing secure file structures
- Managing ownership and privileges

4. Memory & Resource Monitoring

- Observing real-time memory usage
- Analyzing CPU load
- Detecting resource starvation

5. Inter-Process Communication

- Implementing Pipes
- Shared memory concepts
- Basic thread programming

These lab exercises ensure real-world system administration readiness.